

Fermat's Last Theorem

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Introduction

Pierre de Fermat (1601-1665) was a lawyer, a member of the Parliament in Toulouse, and a mathematician. His conjecture is easy to explain: he said that there are no solutions to the following equation when X , Y , and Z are positive integers, and n is a positive integer greater than two:

$$X^n + Y^n = Z^n$$

Fermat created his conjecture in 1637 when he wrote in a copy of *Arithmetica* that he had a proof that was too large to fit in the margin. In 1995, professors Taylor¹ and Wiles² proved the conjecture.

Limiting the Scope of the Problem to Prime Exponents

Professor Van der Poorten³ shows that it is not necessary to prove Fermat's conjecture for any positive exponent n ; rather, it is sufficient to prove it when the exponent p is prime and the greatest common divisor of X and Y equals 1:

$$X^p + Y^p = Z^p \tag{1}$$

Variables

All variables represent integers, unless defined otherwise.

Change of Variables

These new variables represent the differences among the original variables:

$$i = Z - Y$$

$$k = Z - X$$

$$r = X - i$$

$$h = i + k$$

These definitions imply:

$$X = r + i$$

¹ Richard Taylor and Andrew Wiles, "Ring-theoretic properties of certain Hecke algebras", *Annals of Mathematics*, Vol. 141 (1995) pp 553-572.

² Andrew Wiles, "Modular elliptic curves and Fermat's Last Theorem", *Annals of Mathematics*, Vol. 141 (1995) pp 443-551

³ Alf van der Poorten, "Notes on Fermat's Last Theorem", *Canadian Mathematical Society Series of Monographs and Advanced Texts*, Wiley Interscience, 1996, Page 8, note 1.9

$$Y = r + k$$

$$Z = r + i + k$$

If $p > 2$, then (1) can be written as:

$$(X + Y) \sum_{m=0}^{p-1} (-1)^m X^m Y^{p-m-1} = Z^p$$

This implies that:

$$(2r + i + k) \sum_{m=0}^{p-1} (-1)^m X^m Y^{p-m-1} = (r + i + k)^p$$

$$\sum_{m=0}^{p-1} (-1)^m X^m Y^{p-m-1} = \frac{(r + i + k)^p}{(2r + i + k)}$$

$$\sum_{m=0}^{p-1} (-1)^m X^m Y^{p-m-1} = \frac{(r + h)^p}{(2r + h)}$$

All the terms on the left-hand of this equation are integers. This implies that the fraction on the right hand-side is an integer.

i and k divide r^p

Let

$$\text{zero} = Z^p - (X^p + Y^p)$$

Dividing

$$\frac{\text{zero}}{i}$$

yields a quotient with all integer terms, and a remainder:

$$\frac{r^p}{i}$$

This implies that every prime factor of i is also a prime factor of r . A similar analysis applies to k .

1 = gcd (i, k)

Assume that

$$f = \gcd(i, k)$$

Then:

$$i = f i_2$$

$$k = f k_2$$

$$r = f r_2$$

Then (1) can be written as:

$$(f r_2 + f i_2)^p + (f r_2 + f k_2)^p = (f r_2 + f i_2 + f k_2)^p$$

which violates the constraint that the greatest common divisor of X and Y equals 1.

Polynomial Division

In contrast to integer division, polynomial division is performed “with respect to” a variable within a divisor. This can lead to different representations of quotients and remainders when polynomial division with respect to different variables. We can write a division operator as:

$$\text{numerator} = (\text{denominator quotient}) + \text{remainder}$$

If the numerator divided by the denominator is known to be an integer, then the remainder divided by the denominator is an integer.

$$\sum_{m=0}^{p-1} (-1)^m X^m Y^{p-m-1} = \frac{(r+h)^p}{(2r+h)} = \text{quotient} + \frac{\text{remainder}}{(2r+h)}$$

Polynomial division with respect to h

The remainder after polynomial division with respect to h is r^p . This indicates $(2r+h)$ divides r^p .

Polynomial division with respect to r

After polynomial division with respect to r, we have

$$\text{remainder} = \frac{h^p}{2^p}$$

If h is even, the quotient and remainder are integers. If h is odd, the quotient and remainder are rational fractions.

Solve for r

Based upon the division above:

$$n(2r + i + k) = (r + i + k)^p$$

where n is rational (and not necessarily an integer). Solving for r:

$$r = \frac{(i+k)^p - n(i+k)}{2n}$$

Solve for zero

Insert the new value for r into zero, then solve:

$$2^p \text{ zero} = 0$$

This equation has p^2 solutions, p of which are integers where $i = -k$; and $(p^2 - p)$ solutions which are complex.

Since $1 = \gcd(i, k)$, the only solutions for the p integer variables are:

$$i, k = \pm 1$$

which leads to:

$$0 = \sum_{n=1}^{(p+1)/2} -2 C(p, 2n-1) r^{2n-1}$$

where $C(p,n)$ is the binomial coefficient “p choose n”. The sum of the terms on the right hand side cannot equal 0 unless r equals 0,

If none of the $(p^2 - p)$ complex solutions are rational, then Fermat’s conjecture is correct.

Analysis by Gemini

The Appendix contains an analysis by Gemini which concludes that none of the complex solutions are rational.

Conclusion

If Gemini’s analysis is correct, Fermat’s conjecture is correct.

Appendix

Gemini Pro 3.1 provided the following.

Prompt

given integers p, r, i, k , where
 p is prime and greater than 2,
 none of r, i, k equal to zero,

n is rational and greater than 1

$$r = (((i+k)^p - n(i+k))/(2n))$$

$$\text{Let } Z = r+i+k$$

$$\text{Let } X = r+i$$

$$\text{Let } Y = r+k$$

$$X_p = X^p$$

$$Y_p = Y^p$$

$$Z_p = Z^p$$

$$\text{Let zero} = Z_p - (X_p + Y_p)$$

$$\gcd(i, k) = 1$$

$$\gcd(r+i, r+i+k) = 1$$

$$\gcd(r+k, r+i+k) = 1$$

$(r+i)$ and $(r+k)$ and $(r+i+k)$ all are greater than zero,

Using i as a reference variable, does the following equation

$$\text{zero} == 0$$

Have p^2 solutions?

Are p of these solutions $i = -k$?

Are there any expressions in the other $(p^2 - p)$ solutions that can be shown to be not rational?

Ignore Fermat's Last Theorem

Response

Here is a rigorous, purely algebraic recalculation of the system.

By defining the sum and difference of the variables, we can bypass expanding massive polynomials and directly prove the properties of the roots.

Let $S = i + k$ and $D = i - k$. We can now rewrite your defined variables purely in terms of S , D , and n .

First, substitute S into r :

$$r = \frac{S^p - nS}{2n}$$

Now, express Z , X , and Y by clearing the i and k terms:

- $Z = r + S = \frac{S^p - nS + 2nS}{2n} = \frac{S^p + nS}{2n}$
- Because $2i = S + D$, we have $X = r + i = \frac{S^p - nS + n(S+D)}{2n} = \frac{S^p + nD}{2n}$
- Because $2k = S - D$, we have $Y = r + k = \frac{S^p - nS + n(S-D)}{2n} = \frac{S^p - nD}{2n}$

To evaluate $zero == 0$, we substitute these into $Z^p - (X^p + Y^p) = 0$ and multiply both sides by $(2n)^p$ to clear the denominator. We define this new polynomial as $F(i)$:

$$F(i) = (S^p + nS)^p - \left[(S^p + nD)^p + (S^p - nD)^p \right] = 0$$

1. Does the equation have p^2 solutions?

Yes. To find the number of solutions, we determine the degree of the polynomial $F(i)$ by looking at its highest power of S (since $S = i + k$, the degree in S is the degree in i).

- The leading term of $(S^p + nS)^p$ is $(S^p)^p = S^{p^2}$.
- To find the leading term of the subtracted bracket, we use the Binomial Theorem identity $(A+B)^p + (A-B)^p = 2 \sum_{j=0}^{(p-1)/2} \binom{p}{2j} A^{p-2j} B^{2j}$.

- Setting $A = S^p$ and $B = nD$, the largest term (where $j=0$) is $2(S^p)^p = 2S^{p^2}$.

Subtracting these gives a leading term of $S^{p^2} - 2S^{p^2} = -S^{p^2} = -(i+k)^{p^2}$.

Because the leading coefficient is non-zero (-1) and the highest power is p^2 , the Fundamental Theorem of Algebra guarantees exactly **p^2 complex roots** for S .

2. Are p of these solutions $S_i = -k$?

Yes. To prove this, we must show that S^p factors perfectly out of $F(i)$, which would mean $(i+k)^p = 0$.

- In the first term: $(S^p + nD)^p = S^p(S^{p-1} + n)^p$.
- In the symmetric bracket sum: $2 \sum_{j=0}^{(p-1)/2} \binom{p}{2j} (S^p)^{p-2j} (nD)^{2j}$.

The lowest power of S in the symmetric sum occurs at the maximum index $j = \frac{p-1}{2}$, which yields the term $2p(S^p)^1(nD)^{p-1}$. Because every term in the binomial expansion contains at least $(S^p)^1$, we can factor S^p out of the entire equation:

$$F(i) = S^p \cdot Q(i) = 0$$

Setting $S^p = 0$ yields $S_i = -k$ exactly p times.

(To be absolutely rigorous, we must confirm $Q(i)$ does not contain another root of $S_i = -k$. Evaluating Q at $S=0$ gives $Q(-k) = n^p - 2p(nD)^{p-1}$. Because $p > 2$ is prime, $2p$ is not a perfect power, making $Q(-k) \neq 0$. Thus, the multiplicity is exactly p .)

3. Are there expressions in the other $(p^2 - p)$ solutions that are provably non-rational?

Yes. The remaining $p(p-1)$ solutions are the roots of the quotient polynomial $Q(i) = 0$. Even ignoring Fermat's Last Theorem, we can prove that expressions involving these remaining roots are not rational over $\mathbb{Q}$.

Consider the expression for the product of all remaining roots shifted by k (i.e., the roots $S_m = i_m + k$). By Vieta's formulas, the product of these $p(p-1)$ roots is related to the constant term of $Q(S)$ when $S=0$:

$$C = n^p - 2p(nD)^{p-1}$$

For an expression like $S_m = i_m + k$ to be uniformly rational, the constant term C would have to factor cleanly into rational powers. However, C contains the

coefficient $2p$. Because p is an odd prime, $2p$ can **never** be a perfect $(p-1)$ -th power of an integer.

Therefore, the polynomial $Q(i)$ cannot decompose entirely into linear rational factors. This proves that for the remaining solutions, **the expression $S_m = i_m + k$ (and therefore the root i itself) must be an irrational algebraic number** in the complex plane.